

RADIOLOGY OF PNEUMONIA

Douglas S. Katz, MD, and Ann N. Leung, MD

DIAGNOSTIC EFFECTIVENESS OF RADIOGRAPHS

Although chest radiography is often regarded as the reference standard for the diagnosis of community-acquired pneumonia (CAP),^{1, 9} the reliability of this test is limited by significant interobserver variability in radiographic interpretation.^{1, 54} In a series reported by Albaum et al¹ in which two radiologists independently analyzed the chest radiographs of 282 patients with suspected CAP, only fair interobserver agreement ($\kappa = 0.37$) was achieved on the presence or absence of a parenchymal infiltrate, which was diagnosed by both radiologists in 79.4% and 6% of patients, respectively.

Experience in radiographic interpretation can improve individual performance and interobserver variability.⁵⁴ In a series involving 319 patients with suspected CAP, Melbye et al⁵⁴ found substantial agreement between the interpretations rendered by a radiographic panel and a radiologist ($\kappa = 0.71$) and the panel and a pulmonologist ($\kappa = 0.72$); interobserver agreement was only moderate ($\kappa = 0.50$) when the interpretations of the radiographic panel were compared with those of two radiology residents, however.⁵⁴

Pneumonia has traditionally been divided into lobar, bronchopneumonia, and interstitial types.²⁴ As shown by several studies,^{15, 48, 62} this morphologic classification system is of limited usefulness because the radiographic pattern cannot be used to predict the causative organism or even to reliably differentiate among causative groups of pathogens. In a study by Tew and colleagues,⁶² six radiologists who were blinded to clinical data reviewed the initial chest radiographs of 31 patients with laboratory-proven pneumonia. Radiographic

diagnoses were 67% and 65% accurate for bacterial and viral pneumonias, respectively⁶²; no statistical reliability was found for distinguishing bacterial from nonbacterial pneumonia.⁶²

COMMUNITY-ACQUIRED PNEUMONIA

Pneumococcal Pneumonia

Streptococcus pneumoniae is believed to be the most common cause of CAP.²⁶ Pneumococcal infection classically manifests as a lobar pneumonia in which air bronchograms or aerated bronchi are surrounded by opacified lung involving a lobe or most of a lobe (Fig. 1).^{20, 24} In the modern antibiotic era, however, complete lobar involvement by pneumococci is becoming less common, and the infection most often presents with a bronchopneumonia pattern in which patchy, sometimes confluent air-space disease is seen that may be multifocal or bilateral in distribution (Fig. 2).^{20, 24} Two decades ago, Kantor³⁶ reviewed the radiographic appearance of pneumococcal pneumonia in 40 patients and found an equal prevalence (30%) of lobar and bronchopneumonia patterns. Interstitial (defined as irregular linear) as well as mixed interstitial and bronchopneumonia patterns were also relatively common findings, seen in 23% and 17% of patients, respectively.³⁶ In a more recent review of 107 adults with bacteremic CAP pneumococcal pneumonia, 79 patients had unilateral pneumonia that was lobar or segmental in 47 patients and multilobar in 12 patients; 20 patients had diffuse unilateral air-space disease; and 28 patients had diffuse bilateral abnormalities. Diffuse unilateral or bilateral abnormalities were especially common

From the Department of Radiology, Winthrop-University Hospital, Mineola, New York (DSK); Department of Radiology, State University of New York at Stony Brook, Stony Brook, New York (DSK); and the Division of Thoracic Radiology, Department of Radiology, Stanford University Hospital and School of Medicine, Stanford, California (ANL)

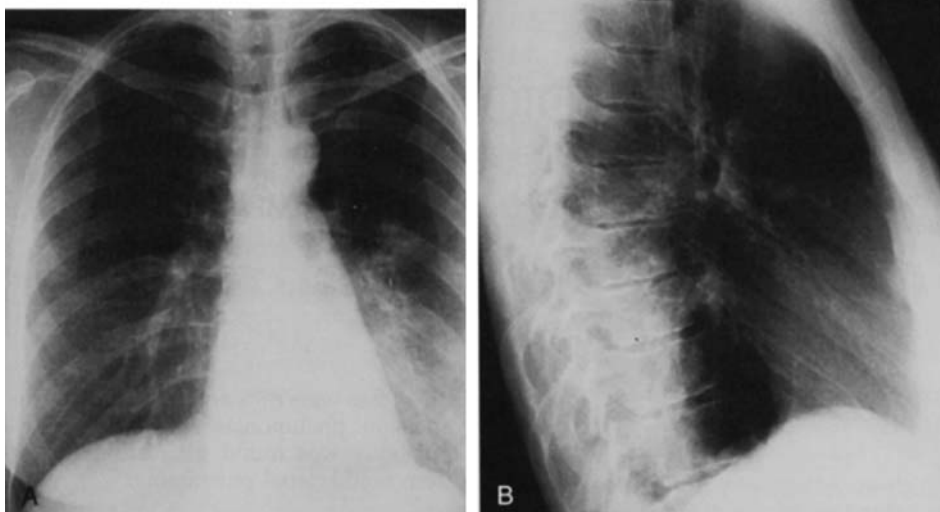


Figure 1. A 39-year-old woman with community-acquired pneumococcal pneumonia. The patient complained of cough and presented with high fever and rust-colored sputum production. PA (A) and lateral (B) radiographs show opacification of a large portion of the posterior left lower lobe.

in patients with underlying chronic obstructive pulmonary disease (COPD). Small- to moderate-sized sterile effusions are commonly associated with pneumococcal pneumonia, with reported prevalences ranging from 46% to 49%.^{61, 66}

Mycoplasma Pneumonia

Reports vary regarding the radiographic appearance of pneumonia caused by *Mycoplasma pneumoniae*. Two main clinical and radiographic groups were identified in a series of 100 patients—one

with an acute clinical presentation who had unilateral or bilateral air-space disease with a lobar or segmental distribution, and the other with longer duration of symptoms and a diffuse bilateral reticulonodular pattern (Fig. 3).⁵⁶ Another series of 60 patients also had variable radiographic appearances, but most commonly confluent or patchy air-space disease was identified.¹⁸ Pleural effusions

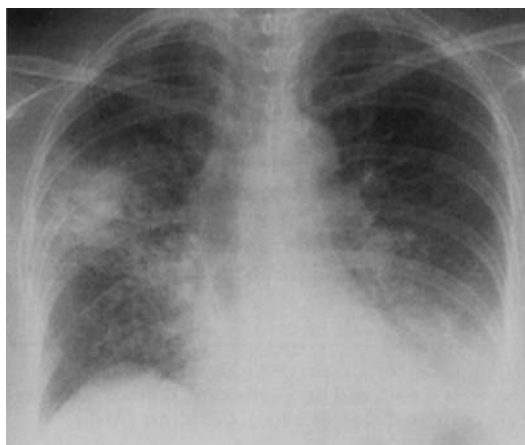


Figure 2. Pneumococcal pneumonia in a 52-year-old woman. Chest radiograph shows multifocal air-space disease associated with a small left-sided pleural effusion.

Rights were not granted to include this figure in electronic media. Please refer to the printed journal.

Figure 3. Mycoplasma pneumonia in a 53-year-old man. Chest radiograph shows diffuse, asymmetric reticulonodular pattern with relative sparing of left upper and middle lung zones. (From Leung AN: Acute shortness of breath. In Jeffrey RB Jr, Ralls PW, Leung AN, et al (eds): Emergency Imaging. Philadelphia, Lippincott Williams & Wilkins, in press; with permission.)

have been reported in mycoplasma pneumonia but are usually small.⁴⁸

Viral Pneumonia

The typical radiographic findings of a viral pneumonia consist of a reticular or reticulonodular pattern that pathologically represents a hemorrhagic pulmonary edema, involving the bronchioles and interlobular septae.²⁴ Often, significant focal or diffuse air-space disease can be found, alone or in combination with the interstitial pattern.^{24, 31} Pleural effusions may also be present,²⁴ although they are usually not the dominant radiographic feature.

Influenza pneumonia can be localized or diffuse, unilateral or bilateral.³¹ Superimposed bacterial infection, particularly with *Staphylococcus aureus*, is common.

Varicella pneumonia may present as patchy, bilateral air-space disease. Clearance is slow, taking up to several months, and, rarely, residual scattered punctate calcifications occur.³¹

Tuberculosis

The radiographic manifestations of tuberculosis (TB) depend on several host factors, including prior exposure to *Mycobacterium tuberculosis* and underlying immune status. In immunocompetent adults with primary TB, segmental or lobar consolidation typically occurs in association with hilar or mediastinal lymphadenopathy (Fig. 4)⁶⁹; multifocal parenchymal involvement occurs in fewer than one quarter of primary cases.^{52, 69} On contrast-en-

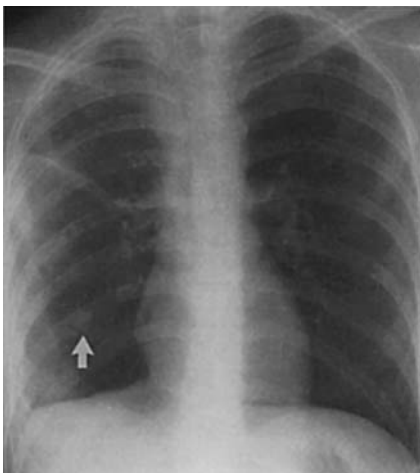


Figure 4. Primary tuberculosis in a 27-year-old Asian woman. Chest radiograph shows a 1-cm nodule in the right lower lung zone (arrow) associated with right paratracheal lymphadenopathy and a loculated right pleural effusion.

hanced CT, tuberculous lymphadenitis may have a characteristic appearance, consisting of central low attenuation associated with peripheral rim enhancement.²⁹ Pleural effusion occurs in approximately one third of affected adults with primary TB and is typically unilateral in distribution.^{10, 65}

Parenchymal opacities situated in the apical and posterior segments of the upper lobes and superior segment of the lower lobes are the characteristic radiographic features of postprimary TB. Parenchymal involvement occurs in more than one segment in the majority of cases and is commonly associated with cavitation.⁶⁹ Endobronchial spread of infection occurs when an area of caseation necrosis communicates with the bronchial tree. Radiographically, endobronchial spread manifests as multiple, ill-defined 5- to 10-mm nodules distributed in a segmental or lobar distribution typically involving the dependent lung zones³⁰; on high-resolution CT scan, the findings of endobronchial spread consist of 2- to 4-mm centrilobular nodules and sharply margined linear branching opacities.²⁸

Miliary tuberculosis, which results from hematogenous dissemination of infection, may occur in association with either primary or postprimary forms of disease. The characteristic radiographic findings are innumerable, 1- to 3-mm nodules disseminated diffusely through the lung parenchyma with mild basilar predominance.⁴¹

ASPIRATION PNEUMONIA

Aspiration pneumonia is an important cause of morbidity and mortality.⁴² Aspiration of sterile gastric contents alone may cause a chemical pneumonitis.⁴⁷ Patients admitted with anaerobic pulmonary infections usually have poor dentition as well as risk factors for aspiration, including decreased mental status and alcoholism.⁴³ Aspiration of gastric contents colonized by various bacteria is probably one of the most common causes of hospital-acquired pneumonia, although other causes have been implicated.³⁷ Aspiration pneumonia should be considered when the history is appropriate and particularly when there is the rapid development of opacities on the chest radiograph.^{24, 42}

In one series of 60 patients with aspiration pneumonia, there was nothing especially characteristic about the chest radiographs alone because the appearance was extremely variable.⁴² Most commonly, bilateral and multicentric opacities were seen, particularly in the right lung, with a perihilar and basilar distribution (Fig. 5). Atypical patterns were common, however. The extent of pulmonary opacification varied from confluent air-space disease to acinar (5–6 mm, poorly defined) nodules. The radiographic appearance also varied depending on the position of the patient during aspiration, as well as the characteristics of the aspirate.⁴²

Anaerobic thoracic infections are caused by a

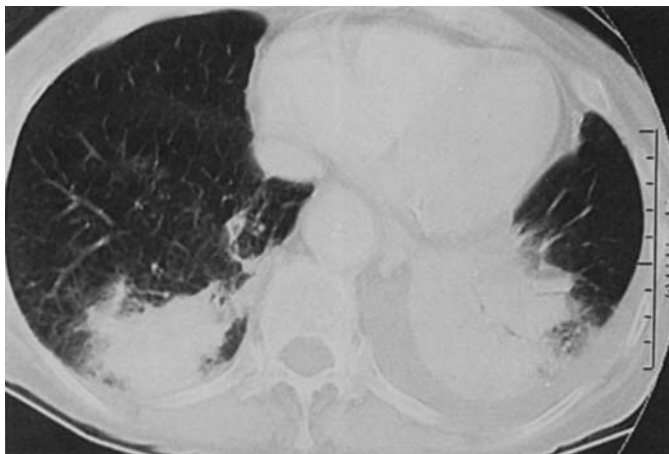


Figure 5. A 72-year-old hospitalized woman with new fever, increasing confusion, and aspiration pneumonia. CT scan shows bibasilar air-space disease and a small left pleural effusion.

variety of organisms that are normally found in the oropharynx but which are abnormally aspirated into the lung, including *Bacteroides*, *Peptococcus*, *Fusobacterium*, and *Propionibacterium*.^{24, 43} Landay⁴³ reviewed the radiographic findings in 69 patients with anaerobic infections. Forty-one were alcoholics or had decreased mental status; 21 patients were hospitalized and had decreased mental status and subsequently developed anaerobic infection. On initial radiographs, infection was confined to the lung in 50%, confined to the pleural space in 30%, and had both pleural and parenchymal components in 20%. The pleural components were empyemas, which usually developed rapidly, whereas the pulmonary component was more indolent in development. The empyemas were large, loculated, and often had air-fluid levels. More than 50% developed areas of necrosis or abscess. Cavities on plain films varied from 2 to 10 cm.⁴³

HOSPITAL-ACQUIRED PNEUMONIA

Gram-negative bacteria and *S. aureus* are the most common causes of nosocomial pneumonia.^{16, 57} A variety of radiographic patterns are seen, including air-space, interstitial, and mixed, although a bronchogenic pattern is often present (Fig. 6). Compounding the problem, it is often difficult to distinguish atelectasis from hospital-acquired pneumonia, and the two may coexist, although atelectasis often develops rapidly and may resolve rapidly.⁵⁷

The incidence of hospital-acquired pneumonia in ventilated patients is as high as 20% in some series, and the mortality rate is high.⁵³ The correct identification of hospital-acquired pneumonia in ventilated patients is a particularly difficult problem, however, and multiple studies have documented the poor accuracy of chest radiography as well as CT in this situation. As many as 60% of mechanically ventilated patients with the clinical

diagnosis of pneumonia have some other process causing abnormalities on their chest radiographs.⁴⁷ The differential diagnosis in these patients includes pulmonary embolism, drug reaction, and adult respiratory distress syndrome (ARDS).⁵³ Even the triad of fever, a new opacity or opacities on the chest radiograph, and purulent tracheal aspirates does not establish the diagnosis of hospital-acquired pneumonia.⁵³ The problem of correctly identifying pneumonia in patients with suspected ARDS is particularly difficult because diffuse abnormalities hide the development of additional processes, and ARDS may be asymmetric on chest radiographs (Fig. 7). Portable chest radiographs are often produced under less-than-ideal technical conditions; usually only an anteroposterior (AP)



Figure 6. Elderly man with hospital-acquired pneumonia following resection of a hepatocellular carcinoma. Chest radiograph shows fairly extensive opacification of the entire right lung. The patient died 2 days later; a specific organism was not identified.

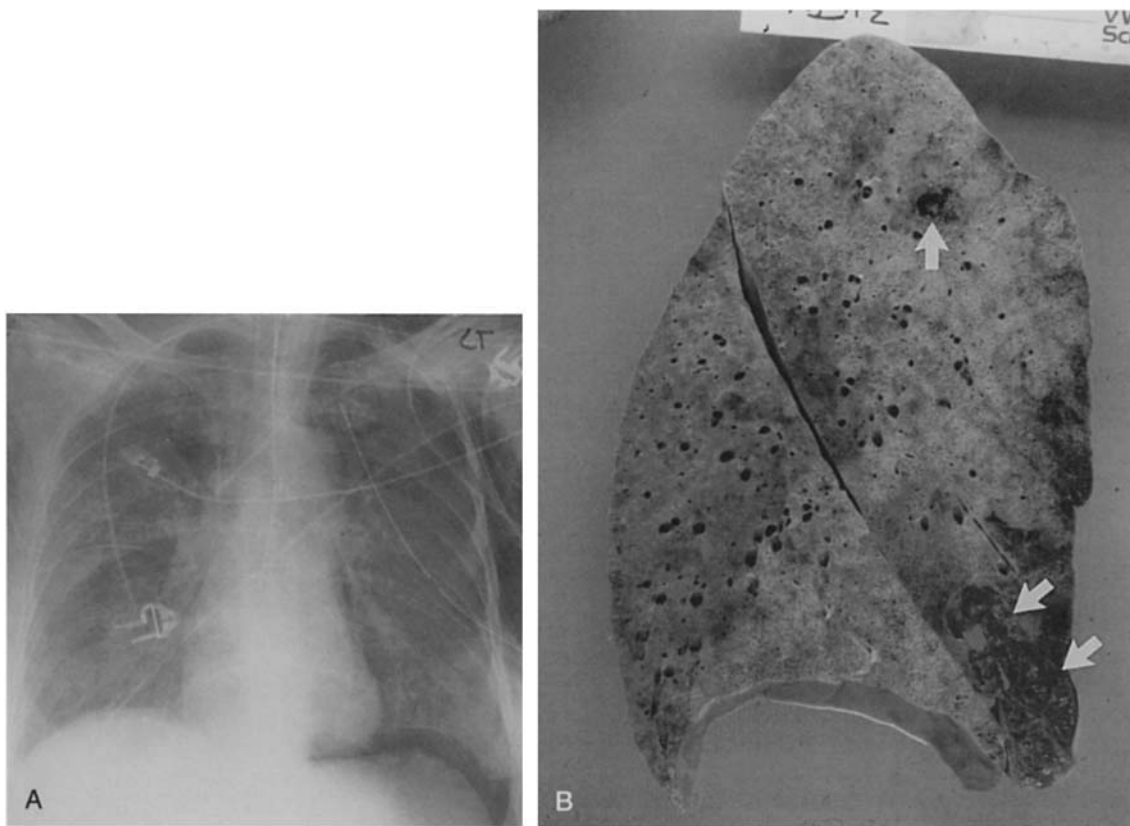


Figure 7. Radiographically occult cavitary lung abscesses in a 65-year-old woman with adult respiratory distress syndrome (ARDS). *A*, Chest radiograph 1 day before death shows diffuse homogeneous opacification of both lungs, associated with a left-sided pneumothorax. *B*, Photograph of the left lung at the time of autopsy demonstrates cavitary abscesses in the lingula and left upper lobe (arrows).

radiograph can be obtained in the intensive care unit (ICU) setting.

Andrews et al² reviewed the autopsy findings in 24 individuals with ARDS. Fourteen had pneumonia histologically but, in five of these patients, pneumonia was not diagnosed clinically prior to death. All 24 had diffuse pulmonary abnormalities on chest radiographs and 11 also had focal opacities. There were no statistical differences with regard to radiographic patterns and various clinical parameters between the patients with ARDS alone and those who had both ARDS and pneumonia.² Similarly, Wunderink et al⁷⁰ reviewed the chest radiographs of 64 mechanically ventilated patients prior to death, including a subgroup of patients with ARDS. Similar difficulties occurred in correctly identifying the presence or absence of pneumonia, particularly when ARDS coexisted.⁷⁰

Lefcoe et al⁴⁴ had two radiologists blindly review the portable chest radiographs of 62 patients in an ICU. Comparison was made with the "gold standard" of cultures obtained during bronchoscopy from protected brush catheter specimens. These cultures were acquired on the same day as the chest films. Interpretations of the chest films were not accurate for the diagnosis of pneumo-

nia—the sensitivity and specificities for the first radiologist for predicting whether the cultures would be positive were 60% and 29%, and for the second radiologist, they were 64% and 27%.⁴⁴

Winer-Muram et al^{67,68} performed two studies in patients on mechanical ventilation. In the first study, the accuracy of three senior thoracic radiologists in predicting the presence of pneumonia in 40 patients based on the portable chest radiographs was only 52% using protected brush catheter specimens obtained at bronchoscopy as the gold standard. In the subgroup of patients with ARDS, the accuracy decreased to 42%. When the radiologists were asked to re-review the chest radiographs and were given the clinical histories, the accuracies did not improve but, in fact, worsened.⁶⁷ In the second study, the accuracy of CT was examined in a group of 31 patients with ARDS.⁶⁸ The diagnostic accuracy of CT for pneumonia was only 69% compared with the cultures from bronchoscopy, and no single CT finding helped accurately establish the diagnosis of pneumonia (including asymmetry of air-space disease, areas of necrosis/cavitation, and air-bronchograms), nor did any combination of these CT findings.⁶⁸

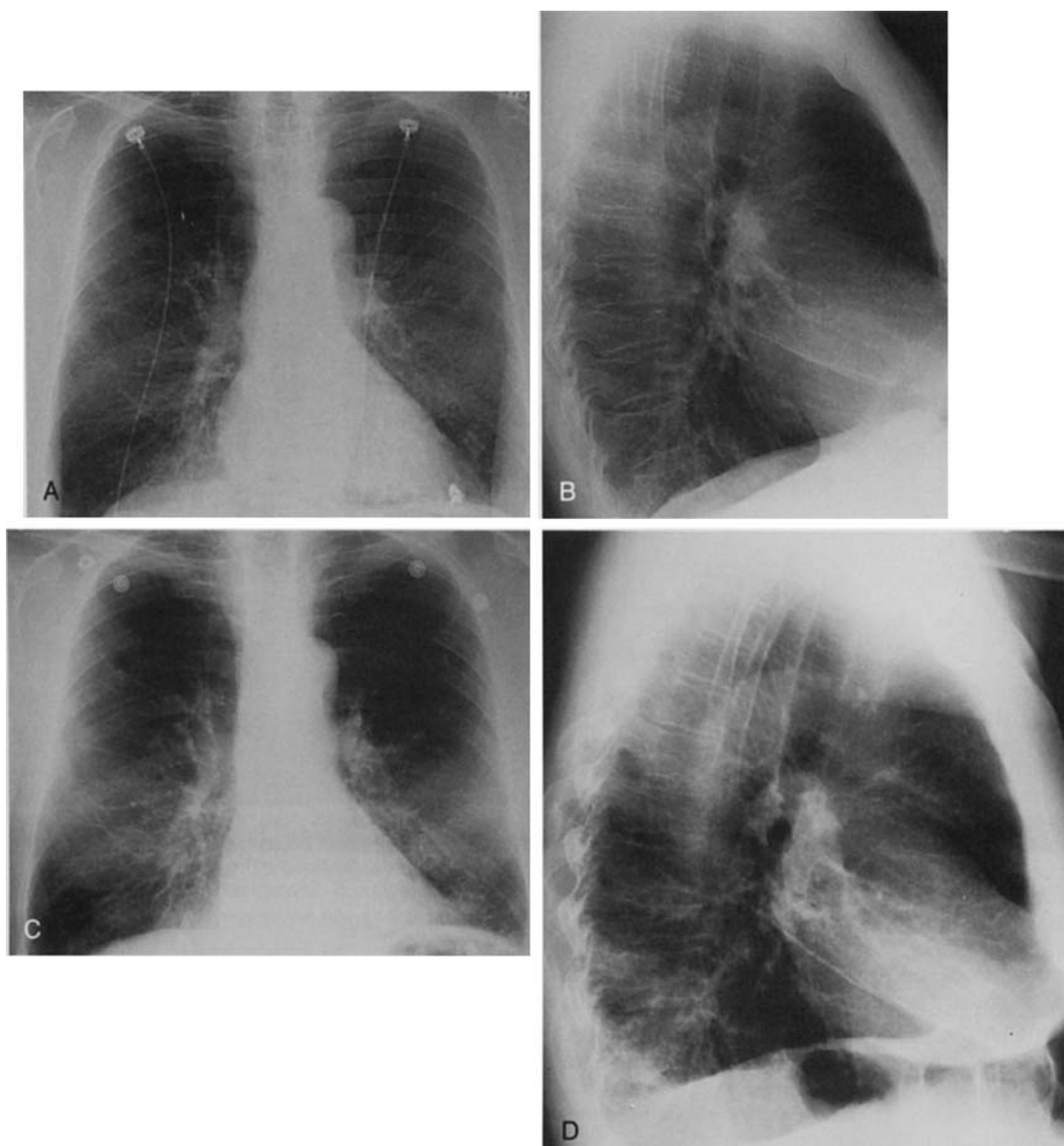


Figure 8. A 72-year-old man with chronic obstructive pulmonary disease (COPD) and delayed radiographic appearance of pneumonia, unrelated to hydration status. PA (A) and lateral (B) chest radiographs on admission show hyperinflated lungs, but no evidence of pneumonia. Patient had new onset of cough and fever, but was not dehydrated on physical examination, and blood pressure and heart rate were normal. Two days later, PA (C) and lateral (D) radiographs reveal opacities in the right middle lobe, the lingula, and the left lower lobe consistent with pneumonia.

ATYPICAL MANIFESTATIONS OF PNEUMONIA

Hydration Status and Radiographic Appearance of Pneumonia

Despite the “conventional wisdom” that has been widely perpetuated for decades that the chest radiograph may be initially normal in a patient with bacterial pneumonia and concurrent moder-

ate to severe dehydration, there is no solid evidence that this phenomenon actually occurs in humans. One canine pneumonia model showed that hydration had no effect on the radiologic or histologic appearance of *S. pneumoniae* pneumonia,⁹ whereas the results of another similar study showed opposite findings.¹¹ There has been only anecdotal evidence of a relationship between dehydration and the delayed appearance of pneumonia on radiographs²³; in fact, one study showed

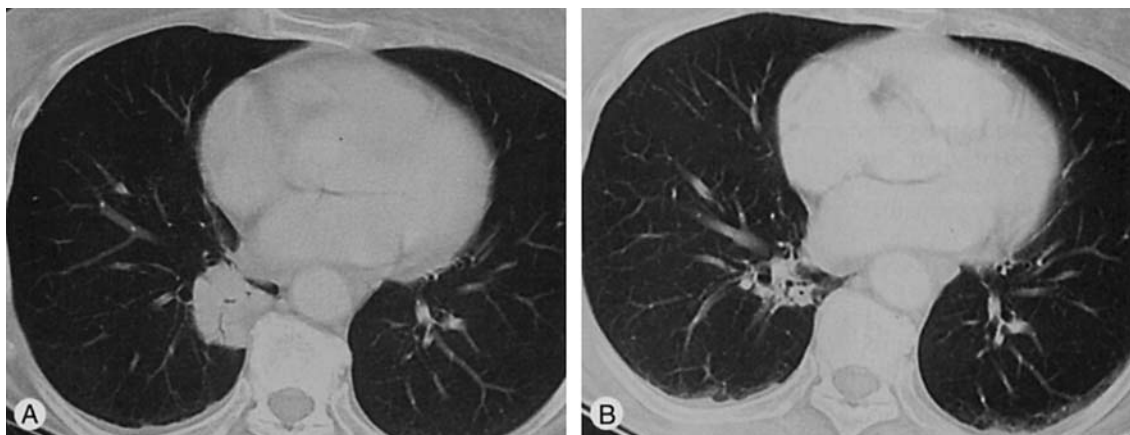


Figure 9. A 67-year-old woman with round pneumonia. A, Initial CT scan shows a round mass with air-bronchograms in the right lower lobe, superiorly. B, Follow-up CT scan several weeks following antibiotic therapy demonstrates near-complete resolution of the mass.

radiographic evidence of pneumonia on admission chest radiographs in 20 dehydrated patients with pneumonia.⁹ The rapid spread of nonaspiration pneumonia is rare but can occur and may be the result of the fast extension of a small focus of infection as opposed to the result of rehydration (Fig. 8).²³

Round Pneumonia

Although more commonly discovered in children, round pneumonia occasionally is seen in adults. As in children, the responsible organism is often *S. pneumoniae*.⁶⁴ Radiographic appearances vary from a small, dense nodule to a large, ill-defined round or oval opacity, the margins of which vary from smooth to lobulated to irregular.⁶⁴ Air bronchograms may also be identified on radiographs or CT scan. Comparison with prior recent radiographs and short-term (3–7 day) interval follow-up are very useful in establishing the correct diagnosis (Fig. 9).^{24, 64}

Pneumonia and Chronic Obstructive Pulmonary Disease

Because of the distortion and destruction of the pulmonary parenchyma that occur in COPD, it is not surprising that patients with COPD who develop pneumonia may have atypical radiographic appearances.¹⁶ Comparison with prior radiographs is most helpful in evaluating these patients (Fig. 10).

Delayed Resolution of Pneumonia and Postobstructive Pneumonia

There are no definitive guidelines as to exactly when follow-up radiographs should be obtained

to document significant improvement or resolution of pneumonia, or when a CT scan or bronchoscopy should be performed to search for underlying or predisposing conditions, particularly lung carcinoma.²⁴ Many studies have documented delayed resolution (i.e., significant residual chest radiographic abnormalities at 4–6 weeks) in a significant percentage of patients with community-acquired



Figure 10. A 63-year-old woman with emphysema who complained of right pleuritic chest pain, fever, and chills. There is patchy opacification of the inferior aspect of the right upper lobe. There is also a focal bulge in the minor fissure. Prior radiographs 1 year earlier demonstrated evidence of COPD but no focal abnormalities were present (not shown). An organism was not recovered on sputum or blood cultures.

as well as hospital-acquired pneumonia.^{24, 47, 49} Factors responsible for delayed clearance of CAP include advanced patient age, underlying lung disease (COPD), alcoholism, and diabetes.⁴⁹ Pneumonias caused by *Legionella* and bacteremic *Pneumococcus* often clear slowly, for unknown reasons.^{14, 38, 48} Anaerobic infections often resolve slowly as well, particularly when necrosis or abscess formation occurs.⁴³ Factors associated with slow resolution of nosocomial pneumonia include underlying pulmonary disease, mechanical ventilation, and aspiration.⁴⁷

Compounding the problem, it has been documented that it is often difficult to correctly identify postobstructive pneumonia on chest radiographs. In one study of 282 patients with suspected CAP, there was poor interobserver reliability in determining whether or not postobstructive pneumonia was present.¹ In addition, the assumption that most postobstructive pneumonias are related to an infectious process is probably not correct. On review of 50 pathology specimens of lung carcinoma,

postobstructive pneumonitis was found to be most likely related to noninfectious processes.⁸ Other processes should be considered in the differential diagnosis of nonresolving air-space disease in addition to postobstructive pneumonitis. These processes include tuberculosis as well as neoplasms such as bronchioloalveolar cell carcinoma and lymphoma (Fig. 11) that produce the air-space disease themselves.

OPPORTUNISTIC INFECTIONS IN IMMUNOCOMPROMISED PATIENTS

Immunocompromised patients are susceptible to a variety of opportunistic pulmonary pathogens. Although chest radiography remains the most widely used imaging technique in the evaluation of the febrile immunosuppressed host, results of radiographs may be normal in up to 10% or more of symptomatic patients. The use of CT allows more sensitive detection of abnormalities in pa-

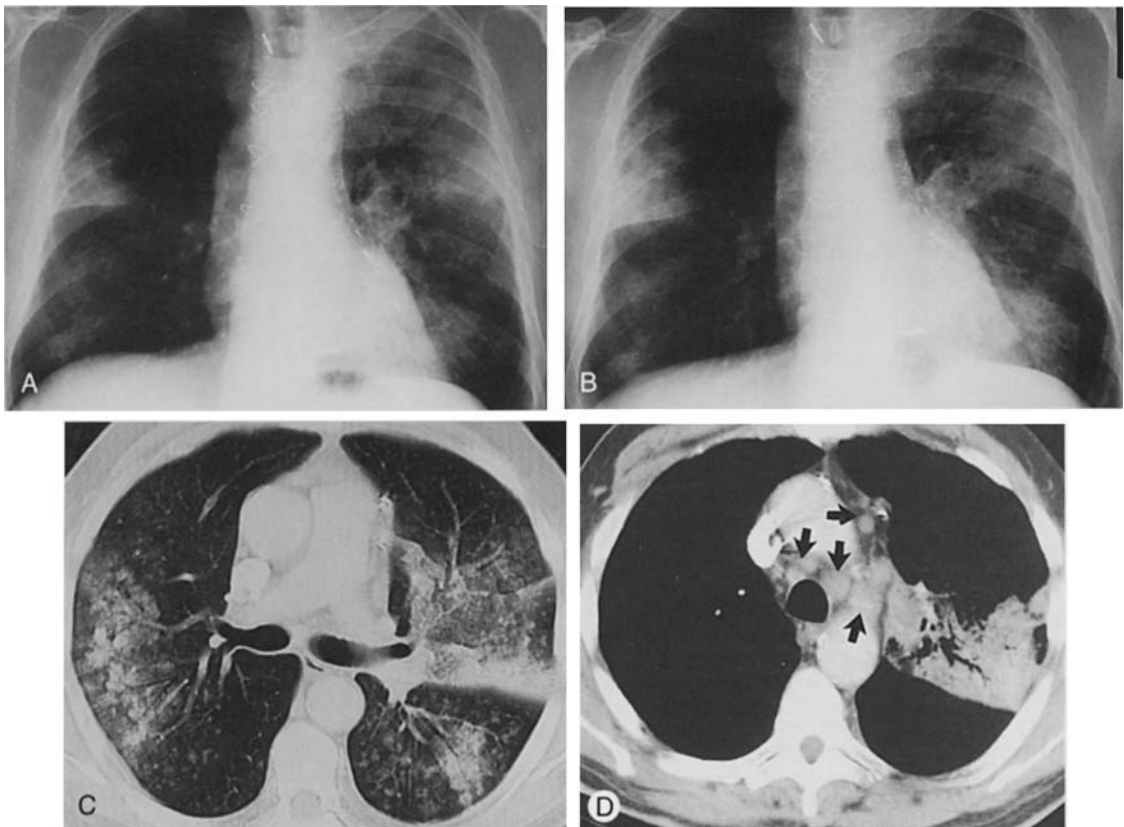


Figure 11. A 76-year-old man status after coronary artery bypass graft in the past with a remote smoking history who complained of shortness of breath and cough. Subacute, nonresolving air-space disease proved to represent bronchioloalveolar cell carcinoma. *A*, Chest radiograph shows patchy bilateral air-space disease. *B*, Follow-up chest radiograph 3 weeks later following therapy with multiple antibiotics shows slight worsening of bilateral air-space disease; the patient had not improved clinically. *C*, CT scan (lung window) shows bilateral air-space disease with multiple small nodular areas. *D*, CT scan (mediastinal window) shows air-space disease in the posterior left upper lobe, and multiple enlarged mediastinal lymph nodes (arrows).

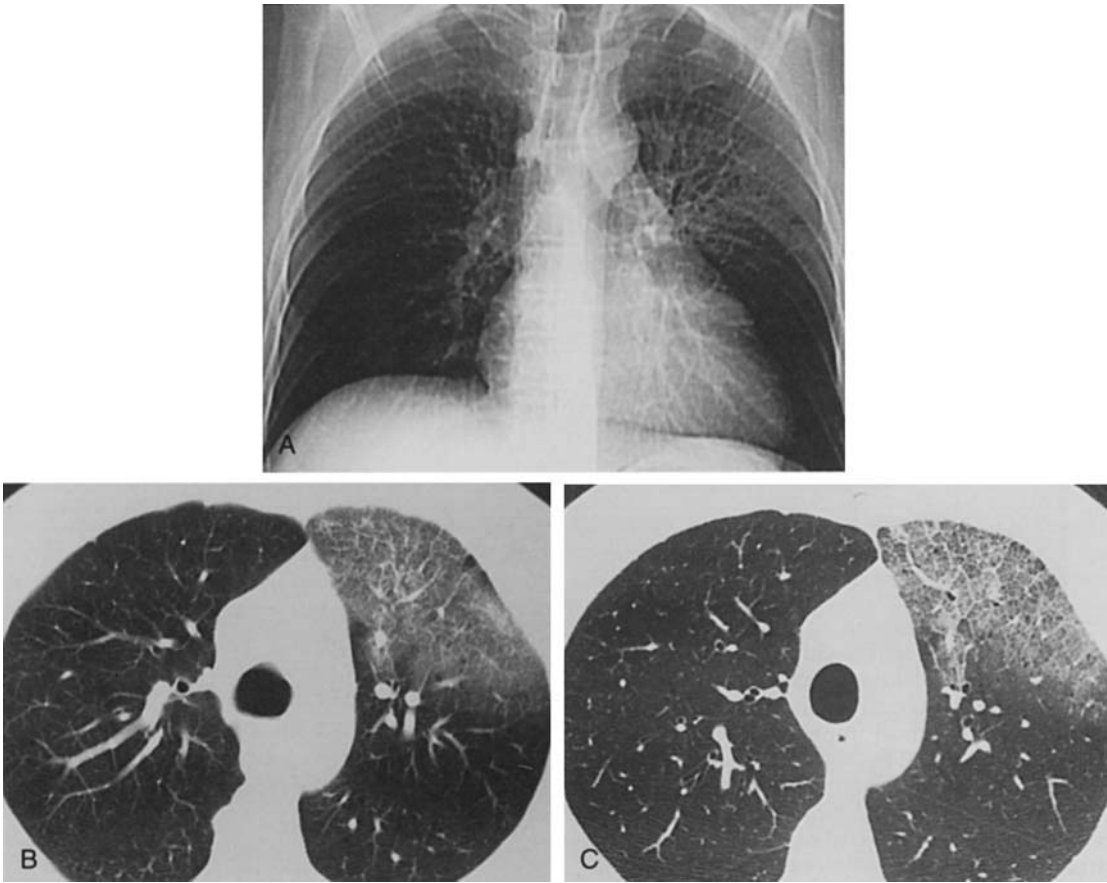


Figure 12. A 49-year-old man with AIDS and pulmonary and central nervous system cryptococcus. Patient developed increasing shortness of breath and mental status changes. *A*, Scout digital radiograph from a CT scan shows increased interstitial markings in the left upper lobe. *B*, CT scan (7 mm thick) shows a combination of ground-glass opacity and increased interstitial markings in the left upper lobe. *C*, High-resolution CT scan (1 mm thick) at a similar level as in *B* reveals fine thickening of the interlobular and intralobular interstitium associated with ground-glass opacity.

tients with normal or equivocal radiographic findings,^{5, 45} may provide a more accurate differential diagnosis,³⁵ and can be used as a guide to the optimal type and site of biopsy (Fig. 12).³³

***Pneumocystis carinii* Pneumonia**

The typical radiographic manifestation of *Pneumocystis carinii* pneumonia (PCP) consists of a diffuse, perihilar reticular or reticulonodular pattern that can progress rapidly to alveolar disease (Figs. 13, 14).¹⁷ Atypical manifestations that have been reported include unilateral or lobar distribution, nodules, cyst formation, pleural effusion, and hilar lymphadenopathy.^{13, 51} Normal radiographs have been reported to occur in 18% to 39% of affected patients with acquired immunodeficiency syndrome.²²

On CT scan, the most characteristic finding of PCP is ground-glass attenuation defined as areas

of increased parenchymal attenuation that do not cause obscuration of the underlying bronchovascular structures.⁵ Distribution of ground-glass opacities is typically bilateral and diffuse; a superimposed reticular pattern and interlobular septal thickening and cystic lesions are occasional associated findings.^{5, 40}

Opportunistic Fungal Infections

Aspergillus is the most common cause of opportunistic fungal infections. In the early stages of the angioinvasive form of disease, an area of consolidation appears that may eventually assume a wedge-shaped pattern as the fungi invade adjacent vessels, leading to thrombosis and hemorrhagic infarction.¹⁹ On CT scan, the angioinvasive stage manifests as a nodule surrounded by ground-glass attenuation (CT halo sign), representing the nidus of fungal infection and adjacent area of hemor-

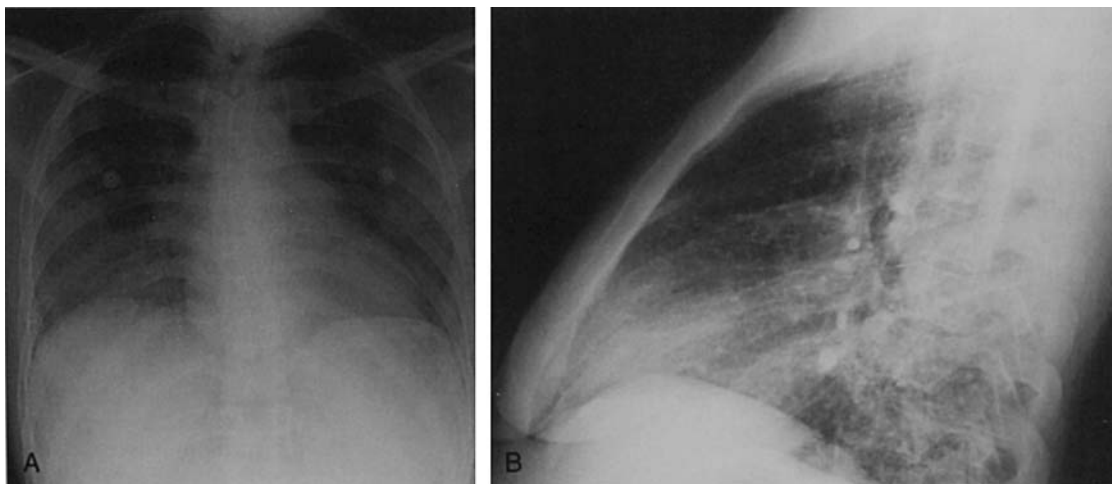


Figure 13. A 30-year-old woman with AIDS and *Pneumocystis carinii* pneumonia. Patient presented with a 3-day history of increasing shortness of breath. PA (A) and lateral (B) radiographs demonstrate low lung volumes and interstitial disease.

rhagic infarction, respectively (Figs. 15, 16).^{27, 39} Cavitation of the nodule (air-crescent sign) occurs in the recovery stage of infection and is usually coincident with clinical improvement and resolution of neutropenia (Fig. 17).

Airway-invasive aspergillosis is much less common than the angioinvasive form of disease. Definite diagnosis requires pathologic demonstration of fungus deep to the basement membrane. Radiographic manifestations of invasive *Aspergillus* of the airways consist of ill-defined 3- to 5-mm nodules or areas of consolidation⁴⁶; on CT scan, consolidation in a peribronchiolar or lobar distribution is seen, often associated with centrilobular nodules.⁴⁶

Rights were not granted to include this figure in electronic media. Please refer to the printed journal.

Figure 14. *Pneumocystis carinii* pneumonia in a 45-year-old man on corticosteroid therapy. High-resolution CT scan shows diffuse ground-glass opacities characteristic of this infection. (From Leung AN: Acute shortness of breath. In Jeffrey RB Jr, Ralls PW, Leung AN, et al (eds): Emergency Imaging. Philadelphia, Lippincott Williams & Wilkins, in press; with permission.)

The radiographic findings of pulmonary candidiasis are variable and range from unilateral segmental or lobar consolidation to bilateral, diffuse, homogeneous or patchy, ill-defined opacities.⁷ On CT scan, a nodular pattern is characteristic^{6, 32}; nodules may be single or multiple, with size varying from 5 to 20 mm.⁶

Cytomegalovirus Pneumonia

Cytomegalovirus (CMV) is the most common viral pathogen responsible for pulmonary infections in organ transplant recipients and patients with AIDS. The radiographic findings have been variably described as consisting of lobar consolida-

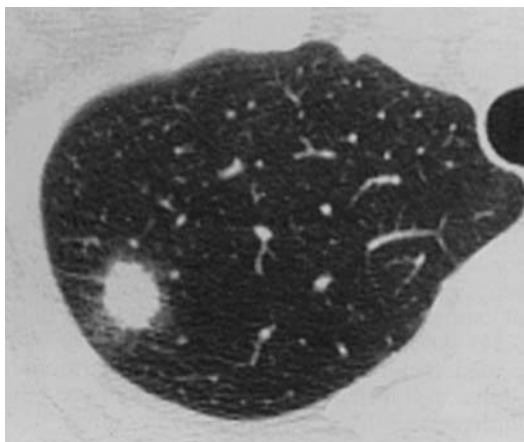


Figure 15. Invasive aspergillosis in a 45-year-old bone marrow transplant recipient. High-resolution CT scan shows a nodule in the right upper lobe, surrounded by ground-glass attenuation (the CT halo sign).

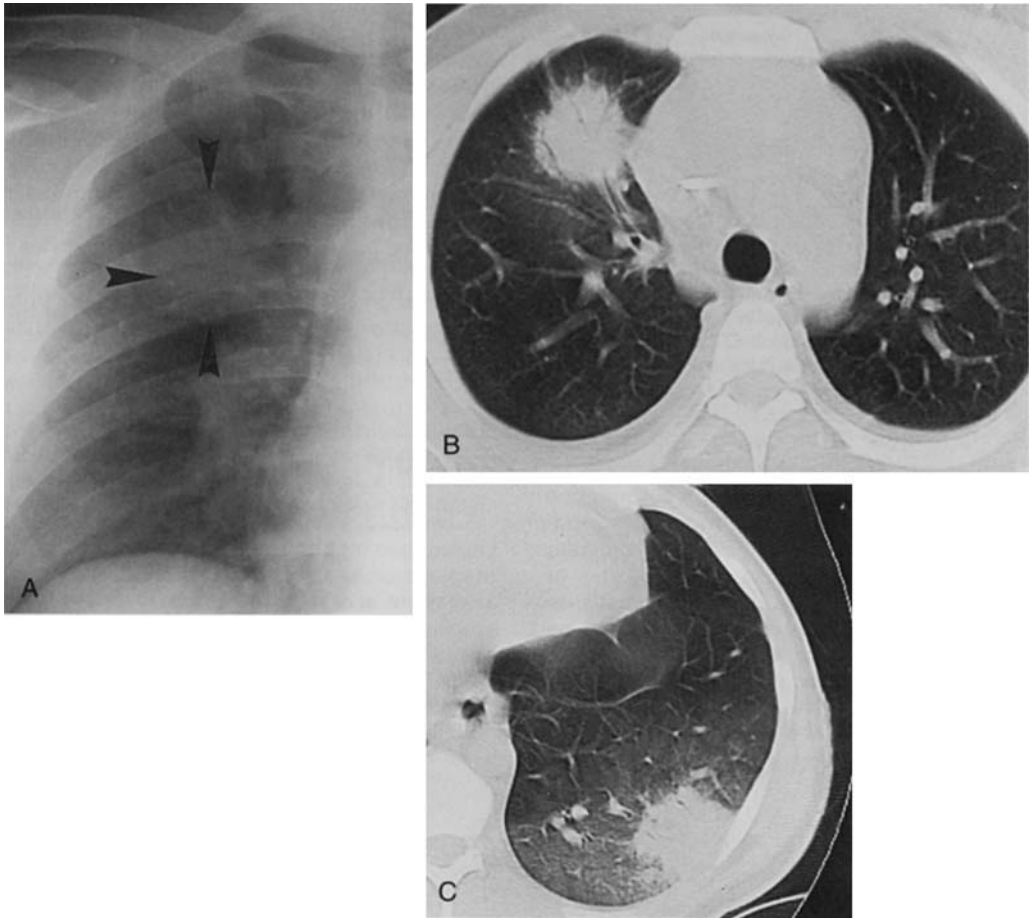


Figure 16. A 31-year-old man with acute myelocytic leukemia and invasive aspergillosis. *A*, Chest radiograph shows a Hickman catheter and a right upper lobe mass that was new, compared with radiographs from 1 week prior. *B–C*, CT scans show an anterior right upper lobe mass which contains air-bronchograms and is surrounded by vague ground-glass opacity (*B*), and a posterior left lower lobe mass with a large area of adjacent ground-glass opacity (CT halo sign) (*C*).

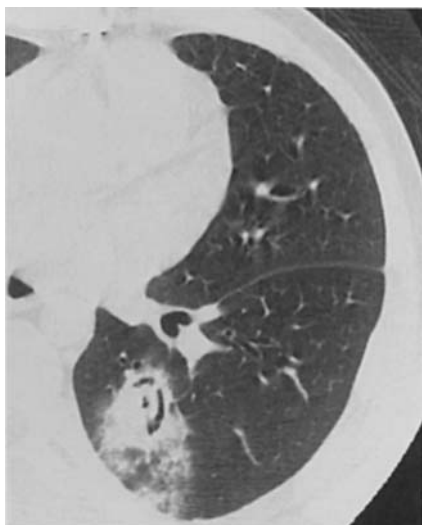


Figure 17. Invasive aspergillosis in a 36-year-old lung transplant recipient. High-resolution CT image shows an ill-defined cavitary mass in the left lower lobe.

tion,⁵⁹ diffuse and focal parenchymal haziness,^{3, 60} and diffuse micronodules (Fig. 18).³⁴ Thin-section CT findings of CMV pneumonia include ground-glass opacity, bronchial wall thickening, reticular opacities, dense consolidation, and nodules or masses.^{34, 50} Normal findings in patients with biopsy-proven CMV pneumonia have also been reported on both radiography⁶⁰ and CT.³⁴

Tuberculosis and Acquired Immunodeficiency Syndrome

The radiographic manifestations of TB in human immunodeficiency virus (HIV)-infected persons depend on the severity of their immunosuppression. Persons with a relatively intact cellular immune function demonstrate radiographic findings similar to the non-HIV-infected population⁶³; at severe levels of immunosuppression, coinfecting patients may have normal radiographs²¹ or demonstrate findings typically associated with primary disease.⁴ In a recent prospective, multi-center study that included 128 HIV-seropositive patients with TB who had both chest radiographs and known CD4 T-lymphocyte levels, a significantly higher prevalence of intrathoracic lymphadenopathy and lower prevalence of cavitation were identified in the patients with CD4 T-lymphocytes fewer than 200/mm³ than the patients with CD4 T-lymphocytes greater than 200/mm³.⁵⁵ Severe immunosuppression in HIV-seropositive patients has also been associated with hematogenous dissemination of TB, resulting in a miliary pattern of disease.⁴

SUMMARY

Chest radiography is the imaging technique of choice in evaluating patients with suspected pneumonia because of its low radiation dose, low cost, and wide accessibility. In daily practice, radiographs are used to confirm the clinical diagnosis of pneumonia, characterize the extent and severity



Figure 18. Cytomegalovirus pneumonia in a 35-year-old bone marrow transplant recipient. Chest radiograph shows a diffuse, bilateral reticulonodular pattern without an associated pleural effusion.

of disease, search for complications such as empyema, monitor the response to therapy, and examine for possible alternative or additional diagnoses.⁵⁸ Although CT scan has no defined role in the routine assessment of patients with either community-acquired or nosocomial pneumonias, its advantages of superior contrast resolution and cross-sectional display can often be helpful in the analysis of complex cases, particularly when radiographic evidence of associated central obstruction, cavitation, lymphadenopathy, or empyema is equivocal.^{12, 58} In the immunocompromised patient population, high-resolution CT has been shown to be more sensitive than plain film radiography in the early detection of pulmonary infections.²⁵

References

- Albaum MN, Hill LC, Murphy M: Interobserver reliability of the chest radiograph in community-acquired pneumonia. *Chest* 110:343, 1996
- Andrews CP, Coalson JJ, Smith JD, et al: Diagnosis of nosocomial bacterial pneumonia in acute, diffuse lung injury. *Chest* 80:254, 1981
- Austin JH, Schulman LL, Mastrobattista JD: Pulmonary infection after cardiac transplantation: Clinical and radiologic correlations. *Radiology* 172:259, 1989
- Batungwanayo J, Taelman H, Dhote R, et al: Pulmonary tuberculosis in Kigali, Rwanda. *Am Rev Respir Dis* 146:53, 1992
- Bergin CJ, Wirth RL, Berry GJ, et al: *Pneumocystis carinii* pneumonia: CT and HRCT observations. *J Comput Assist Tomogr* 14:756, 1990
- Brown MJ, Miller RR, Müller NL: Acute lung disease in the immunocompromised host: CT and pathologic examination findings. *Radiology* 190:247, 1994
- Buff SJ, McLelland R, Gallis HA, et al: *Candida albicans* pneumonia: Radiographic appearance. *AJR Am J Roentgenol* 138:645, 1982
- Burke M, Fraser R: Obstructive pneumonitis: A pathologic and pathogenetic reappraisal. *Radiology* 166:699, 1988
- Caldwell A, Glauser FL, Smith WR, et al: The effects of dehydration on the radiologic and pathologic appearance of experimental canine segmental pneumonia. *Am Rev Respir Dis* 112:651, 1975
- Choyke PL, Sostman HD, Curtis AM, et al: Adult-onset pulmonary tuberculosis. *Radiology* 164:115, 1987
- Cooligan TG, Light R, Duke K, et al: The effect of volume infusion in canine lobar pneumonia. *Am Rev Respir Dis* 121:122, 1980
- Diefenthal HC, Tashjian J: The role of plain films, CT, tomography, ultrasound, and percutaneous needle aspiration in the diagnosis of inflammatory lung disease. *Semin Respir Infect* 3:83, 1988
- Doppman JL, Geelhoed GW, DeVita VT: Atypical radiographic features in *Pneumocystis carinii* pneumonia. *Radiology* 114:39, 1975
- Fairbank JT, Mamourian AC, Dietrich PA, et al: The chest radiograph in Legionnaires' disease: Further observations. *Radiology* 147:33, 1983
- Farr BM, Kaiser DL, Harrison BDW, et al: Prediction of microbial etiology at admission to hospital for pneumonia from the presenting clinical features. *Thorax* 44:1031, 1989
- Fein AM, Feinsilver SH, Niederman MS: Atypical manifestations of pneumonia in the elderly. *Clin Chest Med* 12:319, 1991
- Feinberg SB, Lester RB, Burke B: The roentgen findings in *Pneumocystis carinii* pneumonia. *Radiology* 76:594, 1961
- Finnegan OC, Fowles SJ, White RJ: Radiographic appearances of mycoplasma pneumonia. *Thorax* 36:469, 1981
- Fraser RS: Pulmonary aspergillosis: Pathologic and pathogenetic features. *Pathol Annual* 28:231, 1993
- Genereux GP, Stilwell GA: The acute bacterial pneumonias. *Semin Roentgenol* 25:9, 1980
- Greenberg SD, Frager D, Suster B, et al: Active pulmonary tuberculosis in patients with AIDS: Spectrum of radiographic findings (including a normal appearance). *Radiology* 193:115, 1994
- Gruden JF, Huang L, Turner J, et al: High-resolution CT in the evaluation of clinically suspected *Pneumocystis carinii* pneumonia in AIDS patients with normal, equivocal, or nonspecific radiographic findings. *AJR Am J Roentgenol* 169:967, 1997
- Hall FM, Simon M: Occult pneumonia associated with dehydration: Myth or reality. *AJR Am J Roentgenol* 148:853, 1987
- Heitzman ER: The radiological diagnosis of pneumonia in the adult: A commentary. *Semin Roentgenol* 24:212, 1989
- Heussel CP, Kauczor HU, Heussel G, et al: Early detection of pneumonia in febrile neutropenic patients: Use of thin-section CT. *AJR Am J Roentgenol* 169:1347, 1997
- Hirschtick RE, Glassroth J, Jordan MC, et al: Bacterial pneumonia in persons infected with the human immunodeficiency virus. *N Engl J Med* 333:845, 1995
- Hruban RH, Meziame MA, Zerhouni EA, et al: Radiologic-pathologic correlation of the CT halo sign in invasive pulmonary aspergillosis. *J Comput Assist Tomogr* 11:534, 1987
- Im J, Itoh H, Shim Y, et al: Pulmonary tuberculosis: CT findings—early active disease and sequential change with antituberculous therapy. *Radiology* 186:653, 1993
- Im J, Song KS, Kang HS, et al: Mediastinal tuberculous lymphadenitis: CT manifestations. *Radiology* 164:115, 1987
- Itoh H, Tokunaga S, Asamoto H, et al: Radiologic-pathologic correlations of small lung nodules with special reference to peribronchiolar nodules. *AJR Am J Roentgenol* 130:223, 1978
- Janower ML, Weiss EB: *Mycoplasma*, viral, and rickettsial pneumonia. *Semin Roentgenol* 25:25, 1980
- Janzen DL, Adler BD, Padley SPG, et al: Diagnostic success of bronchoscopic biopsy in immunocompromised patients with acute pulmonary disease: Predictive value of disease distribution as shown in CT. *AJR Am J Roentgenol* 160:21, 1993
- Janzen DL, Padley SPG, Adler BD, et al: Acute pulmonary complications in immunocompromised non-AIDS patients: Comparison of diagnostic accuracy of CT and chest radiology. *Clin Radiol* 47:159, 1993
- Kang EY, Patz EF Jr, Muller NL: Cytomegalovirus pneumonia in transplant patients: CT findings. *J Comput Assist Tomogr* 20:295, 1996
- Kang EY, Staples CA, McGuinness G, et al: Detection and differential diagnosis of pulmonary infections and tumors in patients with AIDS: Value of chest radiography versus CT. *AJR Am J Roentgenol* 166:15, 1996
- Kantor HG: The many radiologic faces of pneumo-

- coccal pneumonia. *AJR Am J Roentgenol* 137:1213, 1981
37. Kollef MH, Schuster DP: Ventilator-associated pneumonia: Clinical considerations. *AJR Am J Roentgenol* 163:1031, 1994
 38. Kroboth FJ, Yu VL, Reddy SC, et al: Clinico-radiographic correlation with the extent of Legionnaire disease. *AJR Am J Roentgenol* 141:263, 1983
 39. Kuhlman JE, Fishman EK, Siegelman SS: Invasive pulmonary aspergillosis in acute leukemia: Characteristic findings on CT, the CT halo sign, and the role of CT in early diagnosis. *Radiology* 157:611, 1985
 40. Kuhlman JE, Kavuru M, Fishman EK, et al: *Pneumocystis carinii* pneumonia: Spectrum of parenchymal CT findings. *Radiology* 175:711, 1990
 41. Kwong JS, Carignan S, Kang E, et al: Miliary tuberculosis: Diagnostic accuracy of chest radiography. *Chest* 110:339, 1996
 42. Landay MJ, Christensen EE, Bynum LJ: Pulmonary manifestations of acute aspiration of gastric contents. *AJR Am J Roentgenol* 131:587, 1978
 43. Landay MJ, Christensen EE, Bynum LJ, et al: Anaerobic pleural and pulmonary infections. *AJR Am J Roentgenol* 134:233, 1980
 44. Lefcoe MS, Fox GA, Leasa DJ, et al: Accuracy of portable chest radiography in the critical care setting: Diagnosis of pneumonia based on quantitative cultures obtained from protected brush catheter. *Chest* 105:885, 1994
 45. Leung AN, Brauner MW, Gamsu G, et al: Pulmonary tuberculosis: Comparison of CT findings in HIV-seropositive and HIV-seronegative patients. *Radiology* 198:687, 1996
 46. Logan PM, Primack SL, Miller RR, et al: Invasive aspergillosis of the airways: Radiographic, CT, and pathologic findings. *Radiology* 193:383, 1994
 47. Lowenkron SE, Niederman MS: Definition and evaluation of the resolution of nosocomial pneumonia. *Semin Respir Infect* 7:271, 1992
 48. MacFarlane JT, Miller AC, Smith WH, et al: Comparative radiographic features of community-acquired legionnaires' disease, pneumococcal pneumonia, mycoplasma pneumonia, and psittacosis. *Thorax* 39:28, 1984
 49. Marrie TJ: Normal resolution of community-acquired pneumonia. *Semin Respir Infect* 7:256, 1992
 50. McGuiness GA, Scholes JV, Garay SM, et al: Cytomegalovirus pneumonitis: Spectrum of parenchymal CT findings with pathologic correlation in 21 AIDS patients. *Radiology* 192:451, 1994
 51. McGuiness GA: Changing trends in the pulmonary manifestations of AIDS. *Radiol Clin North Am* 35:1029, 1997
 52. Medlar G: The behavior of pulmonary tuberculous lesions: A pathologic study. *American Review of Tuberculosis* 71:1, 1955
 53. Meduri GU: Ventilator-associated pneumonia in patients with respiratory failure: A diagnostic approach. *Chest* 97:1208, 1990
 54. Melbye H, Dale K: Interobserver variability in the radiographic diagnosis of adult outpatient pneumonia. *Acta Radiol* 33:79, 1992
 55. Perlman DC, El-Sadr WM, Nelson ET, et al: Variation of chest radiographic patterns in pulmonary tuberculosis by degree of human immunodeficiency virus-related immunosuppression. *Clin Infect Dis* 25:242, 1997
 56. Putnam CE, Curtis AM, Simeone JF, et al: *Mycoplasma pneumoniae*: Clinical and roentgenographic patterns. *Radiology* 124:417, 1975
 57. Rubin SA, Winer-Muram HT, Ellis JV: Diagnostic imaging of pneumonia and its complications in the critically ill patient. *Clin Chest Med* 16:45, 1995
 58. Sagel SS, Slone RM: Lung. In Lee JKT, Sagel SS, Stanley RJ, Heiken JP (eds): *Computed Body Tomography with MRI Correlation*, ed 3. Philadelphia, Lippincott-Raven, 1998, p 442
 59. Schulman LL: Cytomegalovirus pneumonitis and lobar consolidation. *Chest* 91:558, 1987
 60. Shreeniwas R, Schulman LL, Berkmen YM, et al: Opportunistic bronchopulmonary infections after lung transplantation: Clinical and radiographic findings. *Radiology* 200:349, 1996
 61. Taryle DA, Potts DE, Sahn DA: The incidence and clinical correlates of parapneumonic effusions in pneumococcal pneumonia. *Chest* 74:170, 1978
 62. Tew J, Calenoff L, Berlin BS: Bacterial or nosocomial pneumonia: Accuracy of radiographic diagnosis. *Radiology* 124:607, 1977
 63. Theuer CP, Hopewell PC, Elias D, et al: Human immunodeficiency virus infection in tuberculosis patients. *J Infect Dis* 162:8, 1990
 64. Wagner AL, Szabunio M, Hazlett KS, et al: Radiologic manifestations of round pneumonia in adults. *AJR Am J Roentgenol* 170:723, 1998
 65. Wallgren A: The time-table of tuberculosis. *Tubercle* 246, 1948
 66. Wantanakunakorn C, Bailey TA: Adult bacteremic pneumococcal pneumonia in a community teaching hospital, 1992-1996. *Arch Intern Med* 157:1965, 1997
 67. Winer-Muram HT, Rubin SA, Ellis JV, et al: Pneumonia and ARDS in patients receiving mechanical ventilation: Diagnostic accuracy of chest radiography. *Radiology* 188:479, 1993
 68. Winer-Muram HT, Steiner RM, Gurney JW, et al: Ventilator-associated pneumonia in patients with adult respiratory distress syndrome: CT evaluation. *Radiology* 208:193, 1998
 69. Woodring JH, Vandiviere HM, Fried AM, et al: Update: The radiographic features of pulmonary tuberculosis. *AJR Am J Roentgenol* 146:497, 1986
 70. Wunderink RG, Woldenberg LS, Zeiss J, et al: The radiologic diagnosis of autopsy-proven ventilator-associated pneumonia. *Chest* 101:458, 1992

Address reprint requests to

Douglas S. Katz, MD
Department of Radiology
Winthrop-University Hospital
259 First Street
Mineola, NY 11501